

Information Security : TP Clickjacking

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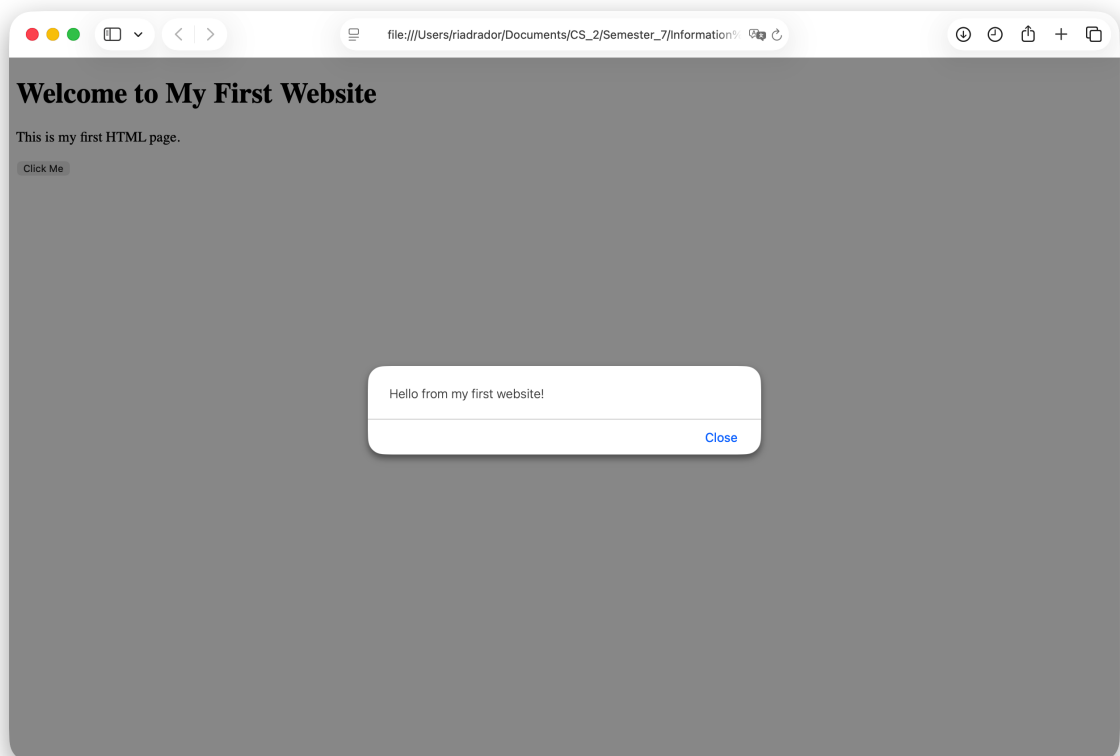
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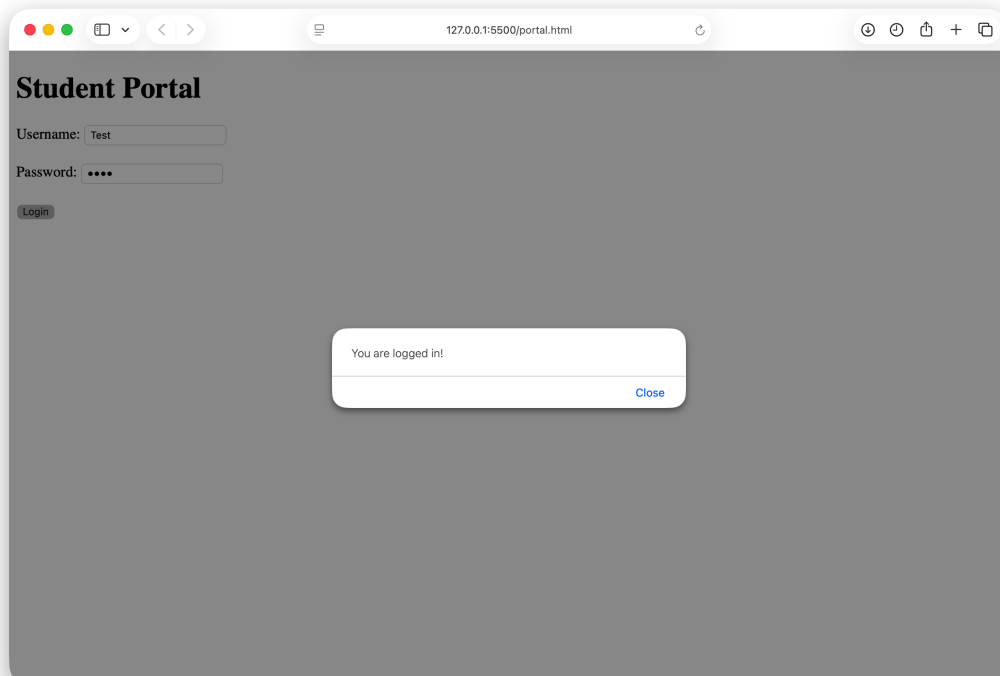
1 HTML Knowledge You Need

```
hello.html x
hello.html > html
1  <!DOCTYPE html>
2  <html>
3  <head>
4  |   <title>My First Website</title>
5  </head>
6  <body>
7  |   <h1>Welcome to My First Website</h1>
8  |   <p>This is my first HTML page.</p>
9
10 |   <button onclick="showMessage()">Click Me</button>
11
12 |   <script>
13 |       function showMessage() {
14 |           alert("Hello from my first website!");
15 |       }
16 |   </script>
17 </body>
18 </html>
```



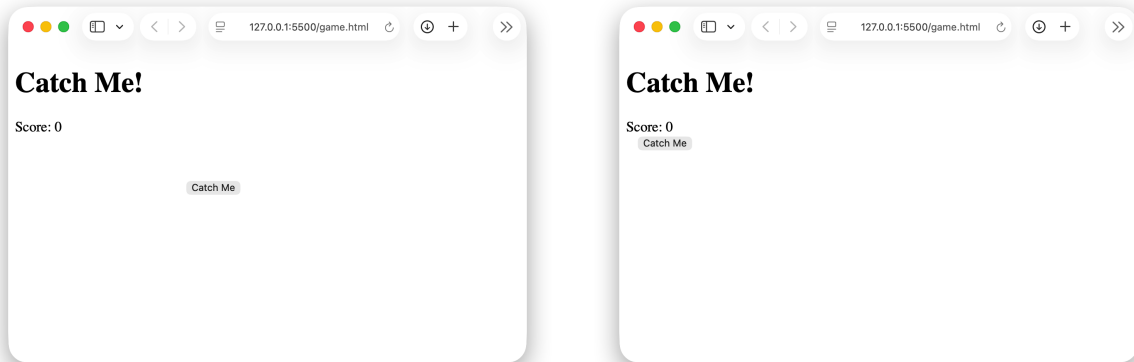
2 Create the Student Portal

```
hello.html x portal.html x
portal.html > html > body > script > login
1  <!DOCTYPE html>
2  <html>
3  <head>
4  |   <title>Student Portal</title>
5  </head>
6  <body>
7  |   <h1>Student Portal</h1>
8
9  |   <!-- Add the name field -->
10 |   <label for="username">Username:</label>
11 |   <input type="text" id="username" placeholder="Enter your name"><br><br>
12
13 |   <!-- Add the password field -->
14 |   <label for="password">Password:</label>
15 |   <input type="password" id="password" placeholder="Enter your password"><br><br>
16
17 |   <!-- Add the Login button -->
18 |   <button onclick="login()">Login</button>
19
20 |   <script>
21 |       function login() {
22 |           // Display "You are logged in!"
23 |           alert("You are logged in!");
24 |       }
25 |   </script>
26 </body>
27 </html>
```



3 Create the Decoy Game

3.1 Task 1

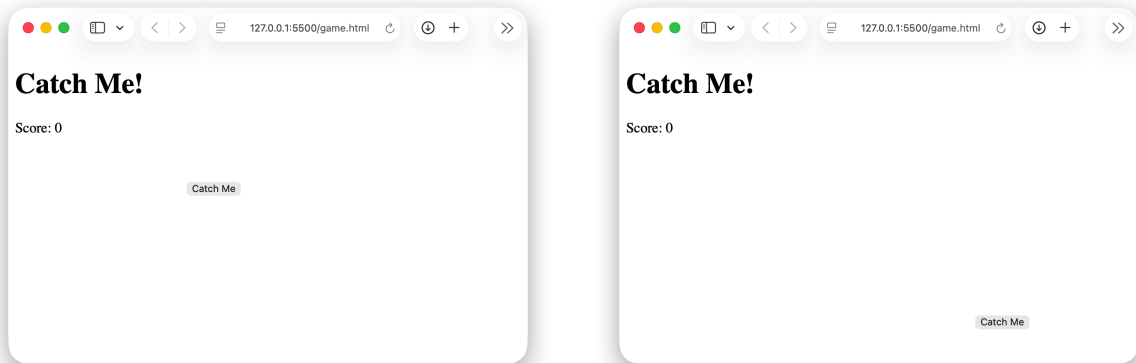


We modify the button position by changing left: and top:

3.2 Task 2

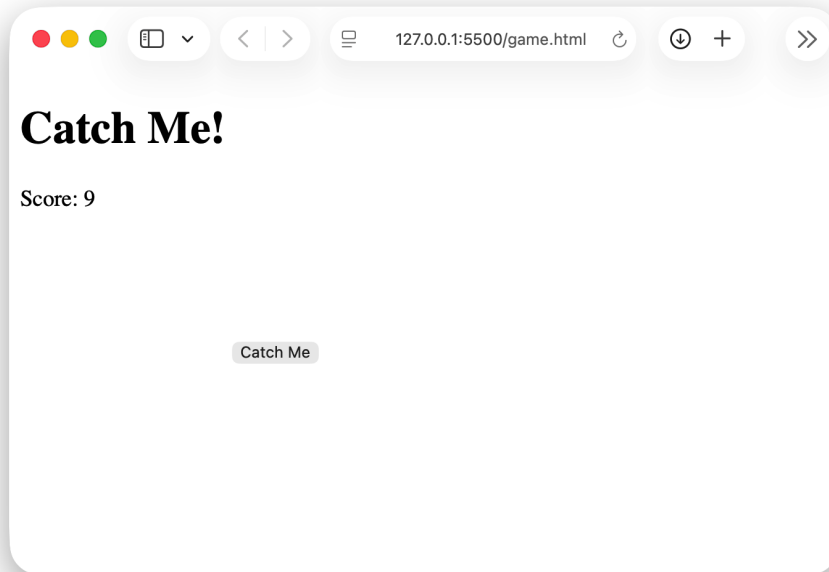
```
<> hello.html x <> portal.html x <> game.html x
<> game.html > html > script > moveButton
1  <!DOCTYPE html>
2  <html>
3      <head>
4          <title>Catch Me!</title>
5          <style>
6              #catchButton {
7                  position: absolute;
8                  top: 150px;
9                  left: 200px;
10             }
11         </style>
12     </head>
13     <body>
14         <h1>Catch Me!</h1>
15         <p>Score: <span id="score">0</span></p>
16         <button id="catchButton" onclick="moveButton()">Catch Me</button>
17     </body>
18
19     <script>
20         function moveButton() {
21             let button = document.getElementById("catchButton");
22             button.style.top = "300px";
23             button.style.left = "400px";
24         }
25     </script>
26 </html>
```

Before and after the click :



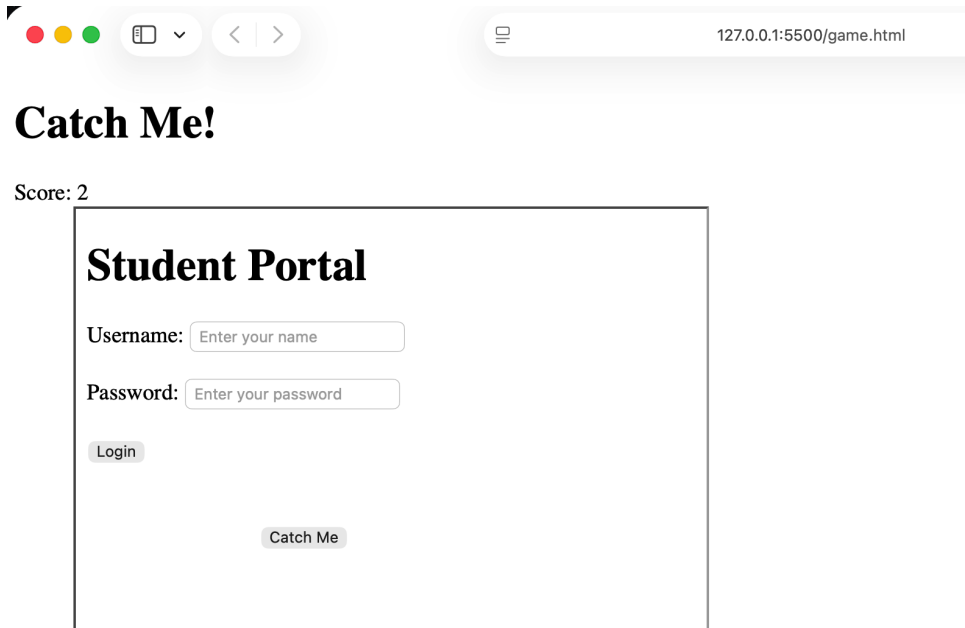
3.3 Task 3

```
<> hello.html X <> portal.html X <> game.html X
<> game.html > html > script > catchMe
1  <!DOCTYPE html>
2  <html>
3      <head>
4          <title>Catch Me!</title>
5          <style>
6              #catchButton {
7                  position: absolute;
8                  top: 150px;
9                  left: 200px;
10             }
11         </style>
12     </head>
13     <body>
14         <h1>Catch Me!</h1>
15         <p>Score: <span id="score">0</span></p>
16         <button id="catchButton" onclick="catchMe()">Catch Me</button>
17     </body>
18
19     <script>
20         let score = 0;
21         let scoreDisplay = document.getElementById("score");
22         let button = document.getElementById("catchButton");
23
24         function catchMe() {
25             score++;
26             scoreDisplay.innerText = score;
27
28             let minX = 50, maxX = 400;
29             let minY = 100, maxY = 350;
30
31             let x = Math.floor(Math.random() * (maxX - minX + 1)) + minX;
32             let y = Math.floor(Math.random() * (maxY - minY + 1)) + minY;
33
34             button.style.left = x + "px";
35             button.style.top = y + "px";
36         }
37     </script>
38 </html>
```



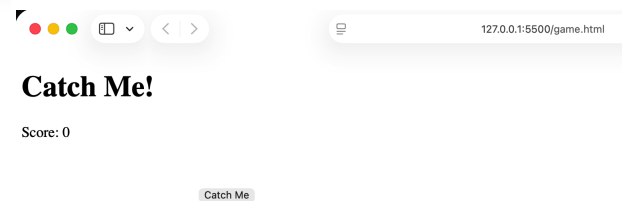
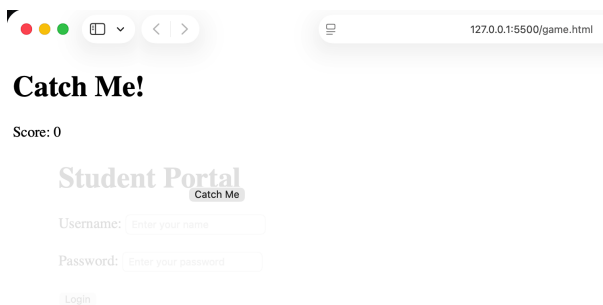
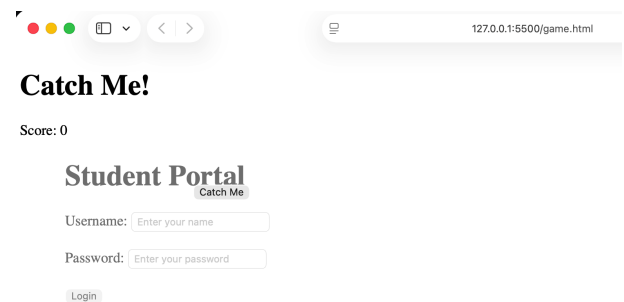
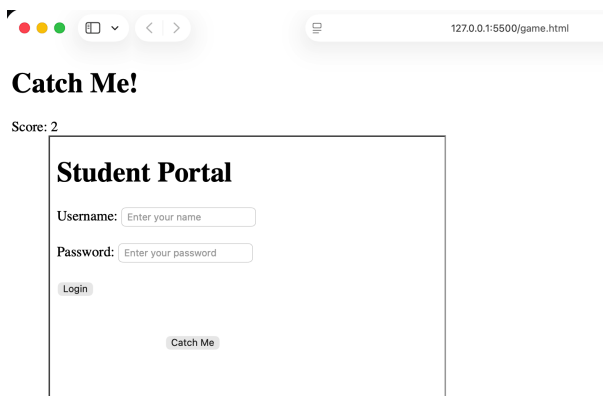
4 About Iframes

```
game.html — Lab 1
hello.html x portal.html x game.html x
game.html > html > head
1 <!DOCTYPE html>
2 <html>
3   <head>
4     <title>Catch Me!</title>
5     <style>
6       #catchButton {
7         position: absolute;
8         top: 150px;
9         left: 200px;
10      }
11
12      #portalFrame {
13        position: absolute;
14        top: 100px;
15        left: 50px;
16        width: 450px;
17        height: 300px;
18      }
19    </style>
20  </head>
21  <body>
22    <iframe id="portalFrame" src="portal.html"></iframe>
23
24    <h1>Catch Me!</h1>
25    <p>Score: <span id="score">0</span></p>
26    <button id="catchButton" onclick="catchMe()">Catch Me</button>
27  </body>
28
29  <script>
30    let score = 0;
31    let scoreDisplay = document.getElementById("score");
32    let button = document.getElementById("catchButton");
33
34    function catchMe() {
35      score++;
36      scoreDisplay.innerText = score;
37
38      let minX = 50, maxX = 400;
39      let minY = 100, maxY = 350;
40
41      let x = Math.floor(Math.random() * (maxX - minX + 1)) + minX;
42      let y = Math.floor(Math.random() * (maxY - minY + 1)) + minY;
43
44      button.style.left = x + "px";
45      button.style.top = y + "px";
46    }
47  </script>
48 </html>
```



5 Hiding the Student Portal

Opacity 1 / 0.5 / 0.1 / 0, I also removed the borders, using `border: none;`



6 The Clickjacking Moment

The image illustrates a clickjacking attack through code and browser screenshots.

HTML Code (game.html):

```
1 <!DOCTYPE html>
2 <html>
3   <head>
4     <title>Catch Me!</title>
5     <style>
6       #catchButton {
7         position: absolute;
8         top: 150px;
9         left: 200px;
10        z-index: 2;
11      }
12
13      #portalFrame {
14        position: absolute;
15        top: 100px;
16        left: 50px;
17        width: 450px;
18        height: 300px;
19        border: none;
20        opacity: 0.5;
21        z-index: 1;
22      }
23    </style>
24  </head>
25  <body>
26    <iframe id="portalFrame" src="portal.html"></iframe>
27
28    <h1>Catch Me!</h1>
29    <p>Score: <span id="score">0</span></p>
30    <button id="catchButton" onclick="catchMe()">Catch Me</button>
31  </body>
```

JavaScript Code:

```
33
34 let score = 0;
35 let scoreDisplay = document.getElementById("score");
36 let button = document.getElementById("catchButton");
37 let frame = document.getElementById("portalFrame");
38
39 function catchMe() {
40   score++;
41   scoreDisplay.innerText = score;
42
43   if (score === 4) {
44     button.style.top = "265px";
45     button.style.left = "60px";
46     frame.style.zIndex = "3";
47   } else {
48     let minX = 50, maxX = 400;
49     let minY = 100, maxY = 350;
50
51     let x = Math.floor(Math.random() * (maxX - minX + 1)) + minX;
52     let y = Math.floor(Math.random() * (maxY - minY + 1)) + minY;
53
54     button.style.left = x + "px";
55     button.style.top = y + "px";
56   }
57 }
58
59 </script>
60 </html>
```

Browser Screenshots:

The left screenshot shows the initial state of the "Catch Me!" game. The score is 4, and the "Student Portal" is visible through the semi-transparent "portalFrame". A "Catch Me" button is positioned over the portal. A message box says "You are logged in!".

The right screenshot shows the result of clicking the button. The button has moved to a new position (60px left, 265px top), and the "portalFrame" is now on top (z-index 3), effectively trapping the user's click on the portal instead of the button.